FINAL

Fiesta Gifts: Mending the Spending

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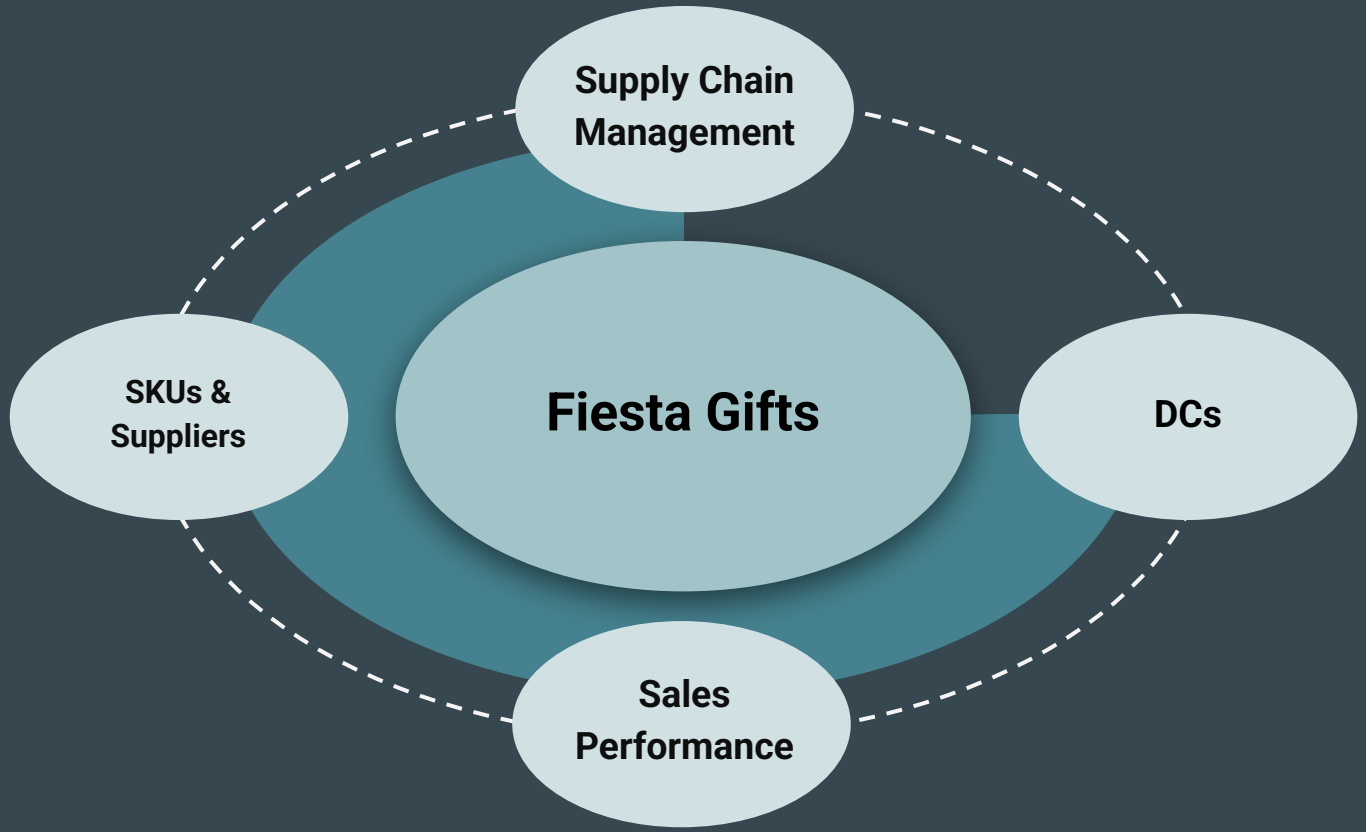
Business Background

Fiesta Gifts (2010~now)

An online retail platform

- 4,500 orders in its first year
- Managing over 3,140 SKUs from 570 suppliers
- Across 20 distribution centers (DCs)

Business Problem



1. Improve Supply Chain Management

Linear Programming

Linear Programming -Pivot table

We aim to analyze the products from 6 key dimensions;

Utilizing the data in 2011.

Year	2011				
SupplierID	Sum of TotalPrice	Average of UnitPrice	Sum of Quantity	Average of SuppDCDist	Average of DCCustDist
AL	21278.152	4.338138547	9643	1270.058402	396.5583128
1001204	5580.358	2.928037313	4083	1138.028172	397.4625
1001305	8211.756	4.389881119	2494	1487.831923	371.0567133
1004201	5658.394	6.755696682	1230	1103.003934	439.6353555
10010303	228.06	1.792608696	206	1231.877391	383.3630435
10012403	1138.354	4.271382716	809	1228.908889	387.0593827
10018101	13.93	13.93	1	1405.14	413.48
10024803	447.3	0.6328	820	1767.0808	357.2728
AZ	40521.824	3.616289562	28635	2439.948855	403.3085791
CA	689548.678	4.075162017	269030	2913.740609	377.6174632
CT	3675.966	2.043416667	1948	1553.122235	378.0793182
FL	188857.228	3.670152916	96513	1568.924649	404.4789906
IA	77896.266	4.114824537	31432	1391.085072	344.9296847
IL	87531.724	4.352198261	33345	1287.338606	363.4886522
IN	31740.87	3.959015228	15766	1128.343677	373.0227121
KS	90892.536	4.646775	28415	1468.919819	354.7434408
KY	767.872	1.9333125	542	1544.809063	430.0326563
MA	6985.006	4.236526596	4139	1902.810293	342.941383
ME	42226.142	4.759385098	15831	1850.727192	335.5992706
MI	69518.232	3.399863736	38541	1210.875504	374.1255667
MN	89753.58	3.739200483	42468	1715.906067	356.0995048
NC	164802.148	3.625774681	80220	1206.561034	402.8872756
NE	30067.842	3.229376049	14859	1561.251579	373.5256979
NH	22641.276	5.031004525	9229	1606.992975	346.1101471
NJ	16043.202	3.230965517	8474	1200.54814	357.5604557
NY	69567.302	3.376637813	33868	1129.527973	372.6810088
OH	55826.694	3.576385762	26973	1245.397198	379.0643419
OK	8882.972	5.983333333	2335	1892.972857	433.1996296
OR	24543.386	2.73842654	15781	2945.739545	378.3024265
RI	80990.532	3.45504632	33255	1450.315082	307.1620717
SC	83291.502	4.218701085	33915	1073.624048	399.3343471
TN	10455.466	4.087706107	5898	1260.361851	374.3190267
TX	213532.228	3.941581321	85767	1424.206677	398.2675419
UT	15221.822	3.856102564	6789	2917.483558	385.3132212
VA	195422.122	2.759526217	85866	955.8068012	433.315762
WA	182445.998	3.870623936	78330	3620.283138	372.5053923
WI	124541.676	2.852581015	66550	1352.225577	375.5248821
Total	2739470.244	3.778257293	1204357	1882.937615	380.2810474

Linear Programming - Sampled data

	SupplierID	Sum of TotalPrice	Average of UnitPrice	Average of SuppDCDist	Average of DCCustDist
AL	1001204	5580.358	2.928037313	1138.028172	397.4625
AL	1001305	8211.756	4.389881119	1487.831923	371.0567133
AZ	100904	14104.034	2.550445783	2126.838795	396.6045984
AZ	1001903	8071.868	3.808387352	2645.571502	349.146996
CA	100202	33625.76	4.263752889	2462.882507	400.8147911
CA	100302	16125.004	4.179010072	2373.352676	379.6031799
CT	1003704	2047.472	2.068018349	1768.917064	388.6119266
CT	10012002	856.744	1.731295082	1276.063115	379.9443443
FL	100305	43453.368	4.663966457	1421.121111	421.4528721
FL	100402	11308.36	2.403587549	2003.560817	368.3720428

We sampled 10 suppliers from 5 states: AL, AZ, CA, CT, and FL.

From the sample data, we know that in 2011, each supplier only supplied to one state. Therefore, we did not break down the data further to the city level.

Linear Programming - Model

Decision variables: Sell Quantity from suppliers

Let Xn be the number of goods planned to be obtained from suppliers in 2012

For n = 1... 10

Where n = 1: 1001204

n = 2: 1001305

...

	SupplierID	Sum of TotalPrice	Average of UnitPrice	Average of SuppDCDist	Average of DCCustDist	sell quantity		Sum of Quantity in 2011
AL	1001204	5580.358	2.928037313	1138.028172	397.4625	0		4083
AL	1001305	8211.756	4.389881119	1487.831923	371.0567133	6577		2494
AZ	100904	14104.034	2.550445783	2126.838795	396.6045984	0		8339
AZ	1001903	8071.868	3.808387352	2645.571502	349.146996	7568		3513
CA	100202	33625.76	4.263752889	2462.882507	400.8147911	0		12820
CA	100302	16125.004	4.179010072	2373.352676	379.6031799	19773		6956
CT	1003704	2047.472	2.068018349	1768.917064	388.6119266	0		1149
CT	10012002	856.744	1.731295082	1276.063115	379.9443443	0		524
FL	100305	43453.368	4.663966457	1421.121111	421.4528721	28760		20116
FL	100402	11308.36	2.403587549	2003.560817	368.3720428	0		8644

Linear Programming - Model

Constraints:

1. Sell quantity in 2012 <= Sell Quantity in 2011 (Control the cost of procurement)
2. Average of SuppDCDist in 2012 <= Average of SuppDCDist in 2011 (Improve logistics efficiency)
3. Average of DCCustDist in 2012 <= Average of DCCustDist in 2011 (Improve DC efficiency)

Number of sell	AL	Constraint 1:	6577	≤	6577				
	AZ		7568	≤	11852				
	CA		19773	≤	19776				
	CT		0	≤	1673				
	FL		28760	≤	28760				
						Average of SuppDCDist in 2011		Goal In 2012	
	SUM of SuppDCDist	Constraint 2:	117606901.3	≤	123548400		1924.258	1800	
	Average of DCCustDist	Constraint 3:	24709662.75	≤	24709680	Average of DCCustDist in 2011			
							395.828	360	

Linear Programming - Model

Objective function:

Maximize the revenue (Quantity from suppliers * UnitPrice)

	SupplierID	Sum of TotalPrice	Average of UnitPrice	Average of SuppDCDist	Average of DCCustDist	sell quantity		Sum of Quantity in 2011			
AL	1001204	5580.358	2.928037313	1138.028172	397.4625	0		4083			
AL	1001305	8211.756	4.389881119	1487.831923	371.0567133	6577		2494			
AZ	100904	14104.034	2.550445783	2126.838795	396.6045984	0		8339			
AZ	1001903	8071.868	3.808387352	2645.571502	349.146996	7568		3513			
CA	100202	33625.76	4.263752889	2462.882507	400.8147911	0		12820			
CA	100302	16125.004	4.179010072	2373.352676	379.6031799	19773		6956			
CT	1003704	2047.472	2.068018349	1768.917064	388.6119266	0		1149			
CT	10012002	856.744	1.731295082	1276.063115	379.9443443	0		524			
FL	100305	43453.368	4.663966457	1421.121111	421.4528721	28760		20116			
FL	100402	11308.36	2.403587549	2003.560817	368.3720428	0		8644			
								259161.3898			
			Number of sell	AL	Constraint 1:	6577	≤	6577			
				AZ		7568	≤	11852			
				CA		19773	≤	19776			
				CT		0	≤	1673			
				FL		28760	≤	28760			
									Average of SuppDCDist in 2011	Goal In 2012	
				SUM of SuppDCDist	Constraint 2:	117606901.3	≤	123548400	1924.258	1800	
				Average of DCCustDist	Constraint 3:	24709662.75	≤	24709680	Average of DCCustDist in 2011		
									395.828	360	
					objective function						
					274461.37						
					more than 2011						
					15299.98						

Recommendation

- Suppliers with zero sell quantities in model should be assessed for potential issues or given targeted support to improve their performance.
- Encourage suppliers with higher sales volumes to maintain or increase their sales efforts.
- Maintain a balanced portfolio of suppliers with moderate sales and reasonable transportation distances to ensure stability and risk mitigation

2. Suppliers

Goal: Minimize shipping cost by choosing valuable suppliers.

Raw Data

Data grouped by

- DCID
- SKUCode
- SupplierID

Calculate Total Distance - from supplier to customer:

=sum(SuppDCDist,DCCustDist)

	A	B	C	D	E	F	G
1	DCID	SKUCode	SupplierID	Quantity	SuppDCDist	DCCustDist	Total Dist
2	1001	10135	1006501	1	3072.77	799.34	3872.11
3	1001	10135	1006503	10	4069.2	362.47	4431.67
4	1001	16045	1001505	500	3402.21	362.47	3764.68
5	1001	16048	1002004	24	4176.97	502.45	4679.42
6	1001	16237	10015001	90	9794.64	724.94	10519.58
7	1001	16237	10015003	30	2732	362.47	3094.47
8	1001	17003	10024101	2196	10184.56	1490.54	11675.1
9	1001	17003	10024102	3609	17585.34	2861.18	20446.52
10	1001	17003	10024103	3600	14654.45	1950.85	16605.3
11	1001	17003	10024104	720	2732	390.17	3122.17
12	1001	20668	100405	24	957.9	320.99	1278.89
13	1001	20674	1001802	2	2065.53	320.99	2386.52
14	1001	20675	1001803	2	1915.21	320.99	2236.2
15	1001	20676	100201	147	5745.63	1101.33	6846.96
16	1001	20676	100202	144	8197.68	780.34	8978.02
17	1001	20676	100204	72	2065.53	390.17	2455.7
18	1001	20676	100205	144	2674.32	780.34	3454.66
19	1001	20677	1001801	72	2229.92	390.17	2620.09
20	1001	20677	1001802	288	8262.12	1560.68	9822.8
21	1001	20677	1001803	76	3830.42	1300.5	5130.92
22	1001	20677	1001804	144	8197.68	780.34	8978.02
23	1001	20679	10012204	6	1135.96	799.34	1935.3

Model - Determine Valuable Suppliers

Same Product ship to the Same DC from different suppliers.

Variable:

- Purchase quantity
- Purchase from supplier(Binary)

Constraint:

- Total purchase quantity remain the same.
- Total shipping distance shorter

Objective function:

- Sum of Total distance after change

	A	B	C	D	E	F	G
1	DCID	SKUCode	SupplierID	Quantity	SuppDCDist	DCCustDist	Average distan / Qua
8	1001	17003	10024101	2196	10184.56	1490.54	4.637777778
9	1001	17003	10024102	3609	17585.34	2861.18	4.872635079
10	1001	17003	10024103	3600	14654.45	1950.85	4.070680556
11	1001	17003	10024104	720	2732	390.17	3.794444444
47815							
47816							
47817							
47818							
47819							
47820							
47821							
47822	SupplierID	Quantity	Average distan / Quan	Total Distan	Purchase (Binary)		
47823	10024101	0	4.637777778	0	0	0	
47824	10024102	0	4.872635079	0	0	0	
47825	10024103	0	4.070680556	0	0	0	
47826	10024104	10125	3.794444444	38418.75	1	2732	
47827							
47828	Total	10125		38418.75			
47829		=		<=			
47830		10125		45156.35			
47831							
47832							
47833							
47834							
47835	Minimize Total shipping cost		38418.75				
47836							

Recommendation

- Choose only one supplier per product (SKU) to streamline procurement.
- Retain suppliers who offer a broader range of products.
- Focus on maintaining organization and operational efficiency.
- Negotiate better terms with selected suppliers to reduce costs.
- Ensure high-quality standards while optimizing supplier relationships.

3. Underutilized Distribution Centers

Goal: Minimize shipping cost by choosing Distribution Centers.

Raw Data

Data sorted by

1. Custcity
2. DCID

	A	B	C	D	E	F	G
1	DCID	DCCity	DCState	DCCustDist	CustCity	CustState	Quantity
2	1013	Columbus	OH	272.81	Akron	OH	1
3	1013	Columbus	OH	272.81	Akron	OH	3
4	1013	Columbus	OH	272.81	Akron	OH	1
5	1013	Columbus	OH	272.81	Akron	OH	1
6	1013	Columbus	OH	272.81	Akron	OH	1
7	1013	Columbus	OH	272.81	Akron	OH	2
8	1013	Columbus	OH	272.81	Akron	OH	4
9	1013	Columbus	OH	272.81	Akron	OH	3
10	1013	Columbus	OH	272.81	Akron	OH	12
11	1013	Columbus	OH	272.81	Akron	OH	3
12	1013	Columbus	OH	272.81	Akron	OH	6
13	1013	Columbus	OH	272.81	Akron	OH	2
14	1013	Columbus	OH	272.81	Akron	OH	1
15	1013	Columbus	OH	272.81	Akron	OH	1
16	1013	Columbus	OH	272.81	Akron	OH	1
17	1013	Columbus	OH	272.81	Akron	OH	1
18	1013	Columbus	OH	272.81	Akron	OH	1
19	1013	Columbus	OH	272.81	Akron	OH	3
20	1013	Columbus	OH	272.81	Akron	OH	3
21	1013	Columbus	OH	272.81	Akron	OH	3
22	1013	Columbus	OH	272.81	Akron	OH	3
23	1013	Columbus	OH	272.81	Akron	OH	1
24	1013	Columbus	OH	272.81	Akron	OH	1
25	1013	Columbus	OH	272.81	Akron	OH	1
26	1013	Columbus	OH	272.81	Akron	OH	1
27	1013	Columbus	OH	272.81	Akron	OH	1
28	1013	Columbus	OH	272.81	Akron	OH	1
29	1013	Columbus	OH	272.81	Akron	OH	4
30	1013	Columbus	OH	272.81	Akron	OH	1
31	1013	Columbus	OH	272.81	Akron	OH	1
32	1013	Columbus	OH	272.81	Akron	OH	1
33	1013	Columbus	OH	272.81	Akron	OH	1
34	1013	Columbus	OH	272.81	Akron	OH	1
35	1013	Columbus	OH	272.81	Akron	OH	1

Model:
Determine Shortest Distance from DC to Customer City

From DC to Customer City

Variable:

- Shipping Quantity
- DC operate or not(Binary)

Constraint:

- Total quantity shipment remain the same.
- Total shipping distance shorter

Objective function:

Sum of Total distance after change

	A	B	C	D	E	F	G	H
1	DCID	CustCity	TotalDCCustDis	TotalQuantity				
667	1002	Roswell	31609.12	334				
668	1005	Roswell	38813.76	454				
669	1006	Roswell	457838.86	14059				
670	1008	Roswell	1053734	11029				
671	1010	Roswell	86718.24	370				
672	1012	Roswell	303514.92	13921				
673	1018	Roswell	67757.42	361				
852								
853								
854								
855								
856								
857								
858	DCID	CustCity	TotalQuantity	Average Dist per Quantity	Total Distance		Op or not (Binary)	
859	1002	Roswell	0	94.63808383	0	>=	0	0
860	1005	Roswell	0	85.49286344	0	>=	0	0
861	1006	Roswell	0	32.56553524	0	>=	0	0
862	1008	Roswell	0	95.54211624	0	>=	0	0
863	1010	Roswell	0	234.3736216	0	>=	0	0
864	1012	Roswell	40528	21.80266648	883618.4669	>=	1	303515
865	1018	Roswell	0	187.6936842	0	>=	0	0
866								
867	Total		40528		883618.4669			
868			=		<=			
869			40528		2039986.32			
870								

Recommendation

- Close low-volume, High-Cost Centers
- Compare the costs and efficiency of each center against benchmark to identify underperformers
- Optimize based on Total Distance
- Regularly review the distribution and shipping strategies to adapt changes

Conclusion

- Support underperforming suppliers and encourage high performers.
- Choose one supplier per product (SKU) and retain those with broad ranges.
- Maintain a balanced portfolio for stability and negotiate better terms.
- Minimize shipping costs by using the shortest routes while ensuring high quality.

Thank You!